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|             | <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Regular Examination – Summer 2023</b><br><b>Course: T.Y. B. Tech.                      Branch : Civil Engineering                      Semester :VI</b><br><b>Subject Code &amp; Name:    BTCVPE604D Water Power Engineering</b><br><b>Max Marks: 60                      Date:19/07/2023                      Duration: 3 Hr.</b>                                                                                                                                                                     |            |            |
|             | <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly.                                                                                                                                                                                          |            |            |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | (Level/CO) | Marks      |
| <b>Q. 1</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            | <b>12M</b> |
| <b>A)</b>   | <b>Define :load factor, capacity factor, utilization factor, diversity factor, Demand factor</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>CO2</b> | <b>6M</b>  |
| <b>B)</b>   | <b>Enlist types of hydropower plant; explain any one in brief..</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>CO1</b> | <b>6M</b>  |
| <b>C)</b>   | A common load is shared by two stations one being a base load plant with 20MW installed capacity and other being a stand by station with 25MW capacity. the yearly output of stand by station is $10 \times 10^6$ kwh and that of the base load plant is $120 \times 10^6$ kwh. The peak load taken by stand by station is 12MW and this station works for 2500hrs during the year. the base load station takes a peak load of 22.8MW find out:<br>1) Annual load factor for both stations<br>2) Plant use factor for both stations<br>3) Capacity factor for both stations | <b>CO2</b> | <b>6M</b>  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |            |
| <b>Q.2</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            | <b>12M</b> |
| <b>A)</b>   | <b>Define intake, give its type and explain any one.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CO3</b> | <b>6M</b>  |
| <b>B)</b>   | <b>Give types of surge tank. Explain any one in detail.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>CO2</b> | <b>6M</b>  |
| <b>C)</b>   | A penstock with an internal diameter of 1.2 m supplies water at a head equivalent to 17.6 kg/cm <sup>2</sup> . There is a possibility of 20% increase in pressure due to transit condition the design stress and efficiency of the joint may be assume to be 1200kg/cm <sup>2</sup> and 85% respectively. Calculate the approximately wall thickness of penstock.                                                                                                                                                                                                           | <b>CO2</b> | <b>6M</b>  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |            |
| <b>Q. 3</b> | <b>Solve Any One of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            | <b>12M</b> |
| <b>A)</b>   | <b>With the help of a detailed diagram of power plant outline its basic components and major sections.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>CO2</b> | <b>6M</b>  |

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| B)          | Give types of underground power house .explain any one in detail.       | CO2 | 6M         |
| C)          | What are the advantages and disadvantage of underground power house?    | CO3 | 6M         |
|             |                                                                         |     |            |
| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                  |     | <b>12M</b> |
| A)          | Differentiate between impulse and reaction turbine.                     | CO2 | 6M         |
| B)          | Explain draft tube & give its types. Explain any one in detail.         | CO2 | 6M         |
| C)          | Explain with neat sketch governing of impulse turbine.                  | CO2 | 6M         |
|             |                                                                         |     |            |
| <b>Q. 5</b> | <b>Solve Any One of the following.</b>                                  |     | <b>12M</b> |
| A)          | Give the advantages and disadvantage of tidal power plant.              | CO1 | 6M         |
| B)          | Give the advantages and disadvantages of pumped storage plant.          | CO3 | 6M         |
| C)          | Explain the concept of pumped storage hydro power plant with its layout | CO3 | 6M         |
|             | <b>*** End ***</b>                                                      |     |            |